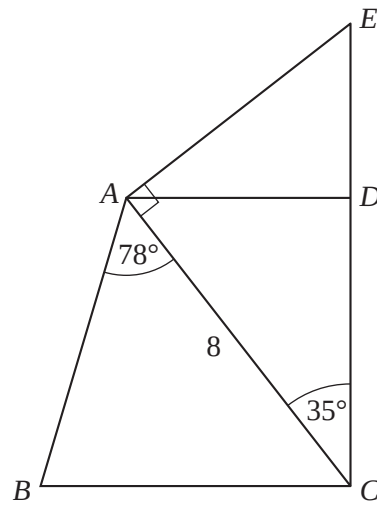


OLEVEL

P2

**A74 TRIG WITH RIGHT
ANGLE TRIANGLES**



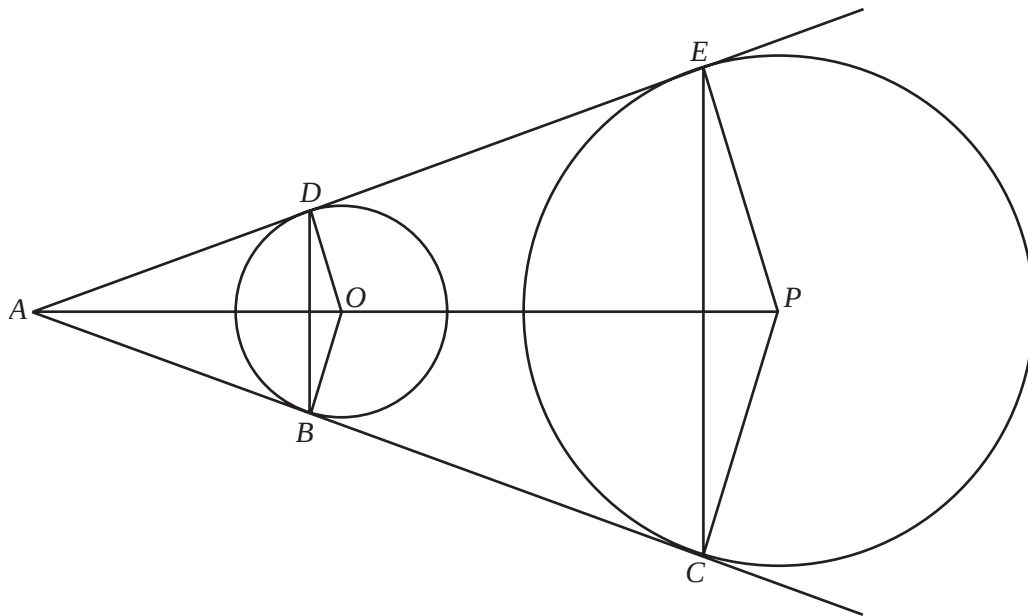
The diagram represents some beams which support part of a roof.

AD and BC are horizontal and CDE is vertical.

$AC = 8$ metres, $\hat{BAC} = 78^\circ$, $\hat{ACD} = 35^\circ$ and $\hat{CAE} = 90^\circ$.

Calculate the length of the beam

- (a) AD , [2]
- (b) CE , [2]
- (c) AB . [3]

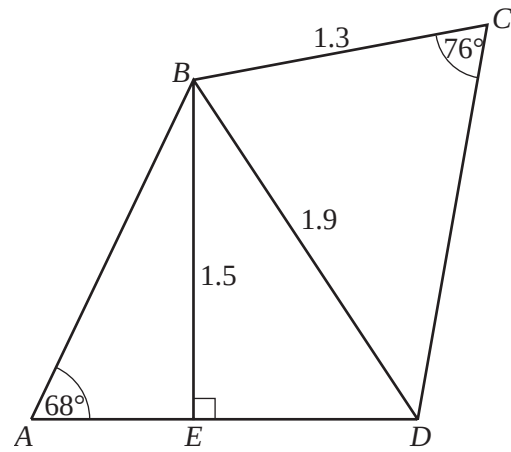


The diagram shows two circles with centres O and P .
 ABC and ADE are tangents to the circles at B , C , D and E as shown.
 AOP is a straight line.

- (a) Giving a reason for your answer, write down angle ABO . [1]
- (b) It is given that $OB = 6$ cm, $AO = 13$ cm and $PC = 15$ cm.
- (i) Show that angle $OAB = 27.5^\circ$, correct to one decimal place. [1]
- (ii) Calculate AC . [2]
- (iii) Calculate CE . [3]

- 3 The diagram represents a framework.

$BC = 1.3$ m, $BD = 1.9$ m and $BE = 1.5$ m.
 $\hat{BCD} = 76^\circ$, $\hat{BAE} = 68^\circ$ and $\hat{BED} = 90^\circ$.

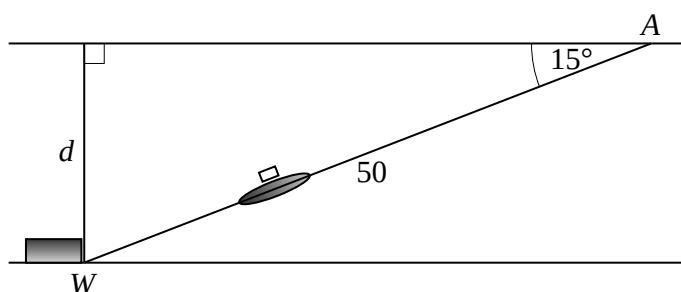


Calculate

- (a) $\hat{DBE}$, [2]
(b) AE , [2]
(c) $\hat{BDC}$. [3]

- 4 A small submarine dived in a straight line from a point A on the surface to examine an object at the point W on the seabed.

(a)

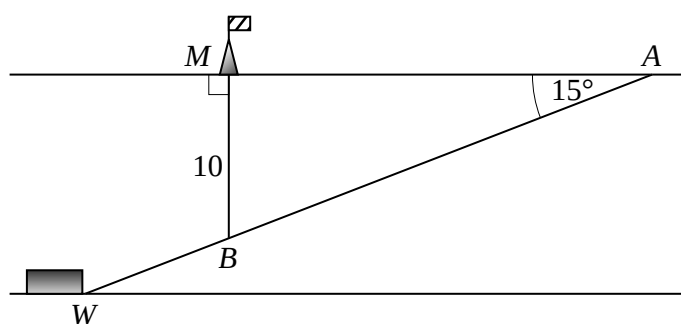


It dived at an angle of 15° to the horizontal and reached W after travelling 50 m.

Calculate the depth, d metres, of the seabed at W .

[2]

(b)



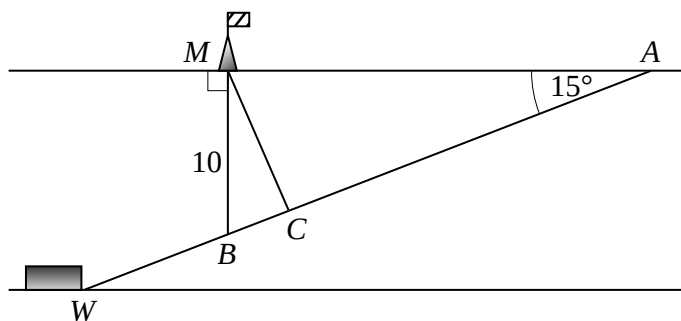
A marker is at the point M on the surface.

When at B , the submarine was 10 m vertically below M .

Calculate AB .

[3]

(c)



When at C , the submarine was at its nearest point to M .

(i) Find $\hat{BMC}$.

[1]

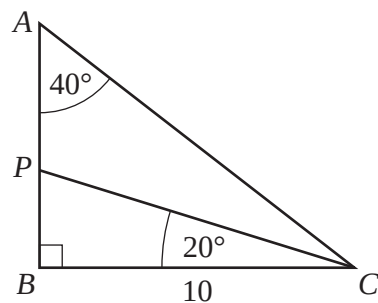
(ii) Calculate CM .

[2]

5

ABC is a triangle in which $\hat{ABC} = 90^\circ$, $\hat{BAC} = 40^\circ$ and $BC = 10$ cm.

P is the point on AB such that $\hat{PCB} = 20^\circ$.



Calculate

(a) PB ,

Answer cm [2]

(b) AP ,

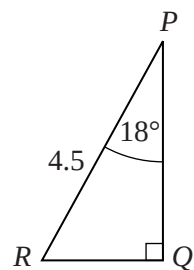
Answer cm [2]

(c) the perimeter of triangle PBC .

Answer cm [3]

- 6 (a) In triangle PQR , $\hat{RPQ} = 18^\circ$, $\hat{PQR} = 90^\circ$ and $PR = 4.5$ m.

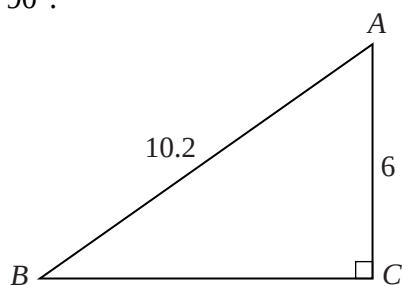
Find PQ .



Answer m [2]

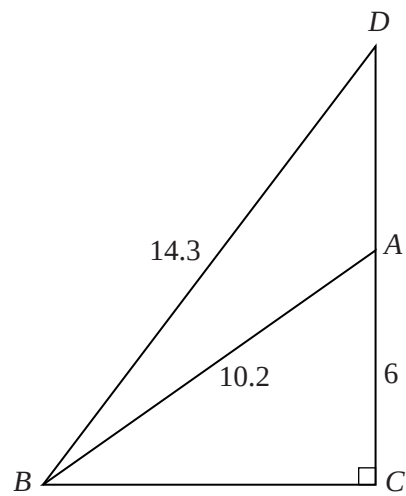
- (b) In triangle ABC , $AB = 10.2$ m, $AC = 6$ m and $\hat{ACB} = 90^\circ$.

(i) Find $\hat{ABC}$.



Answer [2]

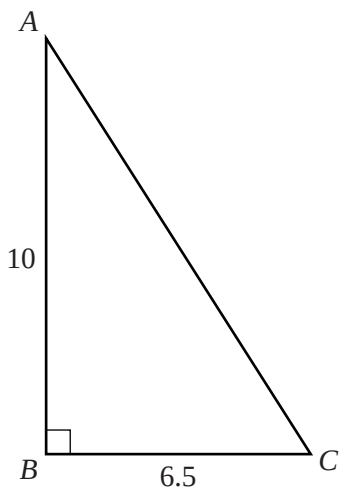
(ii)



D is the point on CA produced such that $BD = 14.3$ m.

Find AD .

Answer m [4]

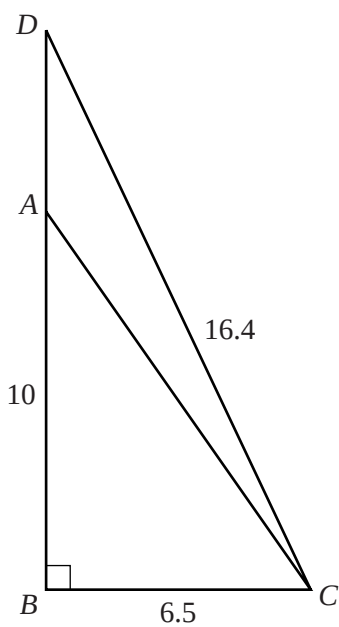


In triangle ABC , $AB = 10$ m, $BC = 6.5$ m and $ABC = 90^\circ$.

(a) Find $\hat{ACB}$.

Answer [2]

(b)



D is the point on BA produced such that $CD = 16.4$ m.

(i) Find AD .

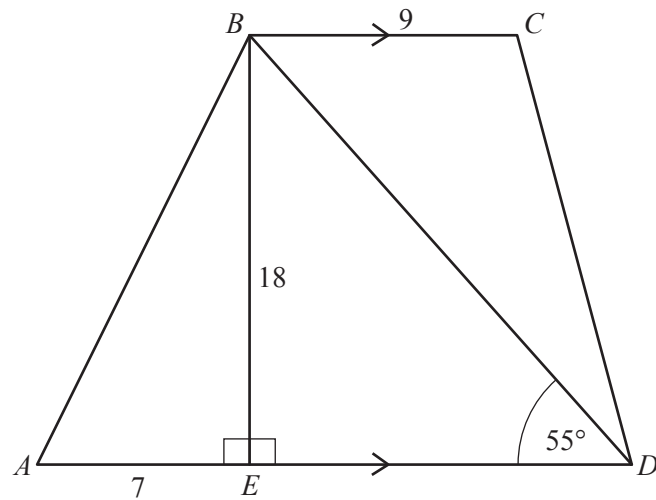
Give your answer in metres and centimetres, correct to the nearest centimetre.

Answer m cm [3]

(ii) Find $\hat{DCB}$.

Answer [2]

- 8 a) $ABCD$ is a trapezium with BC parallel to AD .



E is the point on AD such that BE is perpendicular to AD .
 $\angle BDA = 55^\circ$, $AE = 7$ cm, $BE = 18$ cm and $BC = 9$ cm.

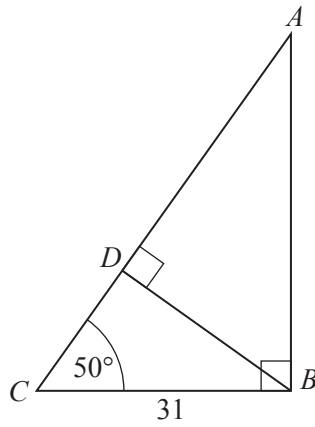
Calculate

- (i) $\angle BAE$,

Answer [2]

- (ii) the area of the trapezium $ABCD$.

Answer cm^2 [4]



In the triangle ABC , $\hat{ABC} = 90^\circ$, $\hat{ACB} = 50^\circ$ and $BC = 31$ m.
 D is the point on AC such that $\hat{BDA} = 90^\circ$.

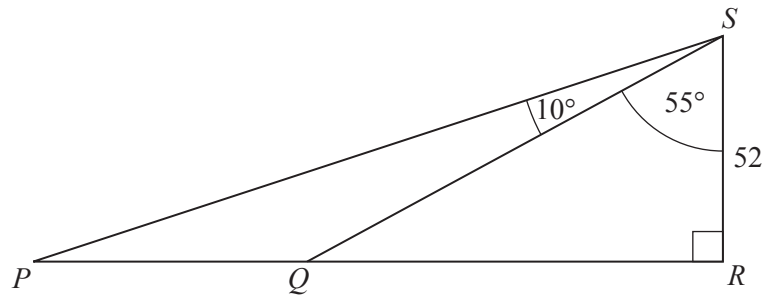
- (i) Show that $CD = 19.93$ m, correct to 2 decimal places.

[2]

- (ii) Calculate AD .

Answerm [3]

(b)



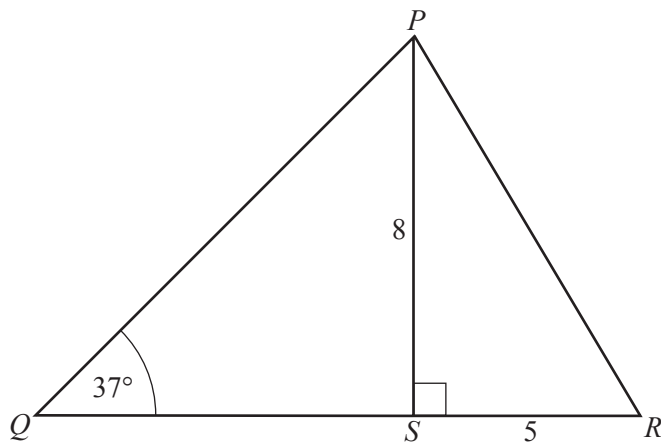
Two boats are at the points P and Q .
 RS is a vertical cliff of height 52 m.
 $\hat{PSQ} = 10^\circ$ and $\hat{QSR} = 55^\circ$.

(i) State the angle of depression of P from S .

Answer[1]

(ii) Calculate the distance, PQ , between the boats.

Answerm [3]



The diagram shows a triangle PQR with $\hat{PQR} = 37^\circ$.

S is the point on QR such that $\hat{PSR} = 90^\circ$, $PS = 8$ cm and $SR = 5$ cm.

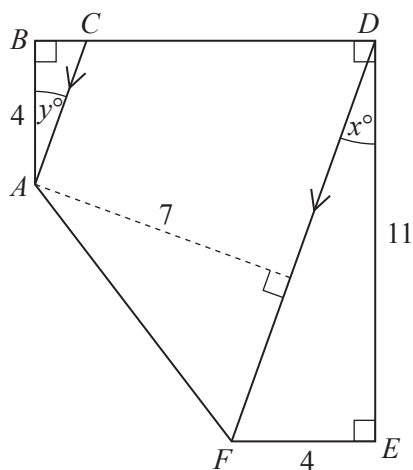
Calculate

(i) PR ,

Answer cm [2]

(ii) the shortest distance from S to PQ .

Answer cm [3]



In the framework $ABCDEF$, BCD is a straight line, and CA is parallel to DF .
 $\hat{ABD}$, $\hat{BDE}$ and $\hat{DEF}$ are right angles.
 $AB = 4$ m, $DE = 11$ m and $EF = 4$ m.

- (a) $\hat{FDE} = x^\circ$.

Show that $x = 20.0$ correct to 3 significant figures.

[2]

- (b) $\hat{BAC} = y^\circ$.

Stating your reasons, explain why $y = x$.

[1]

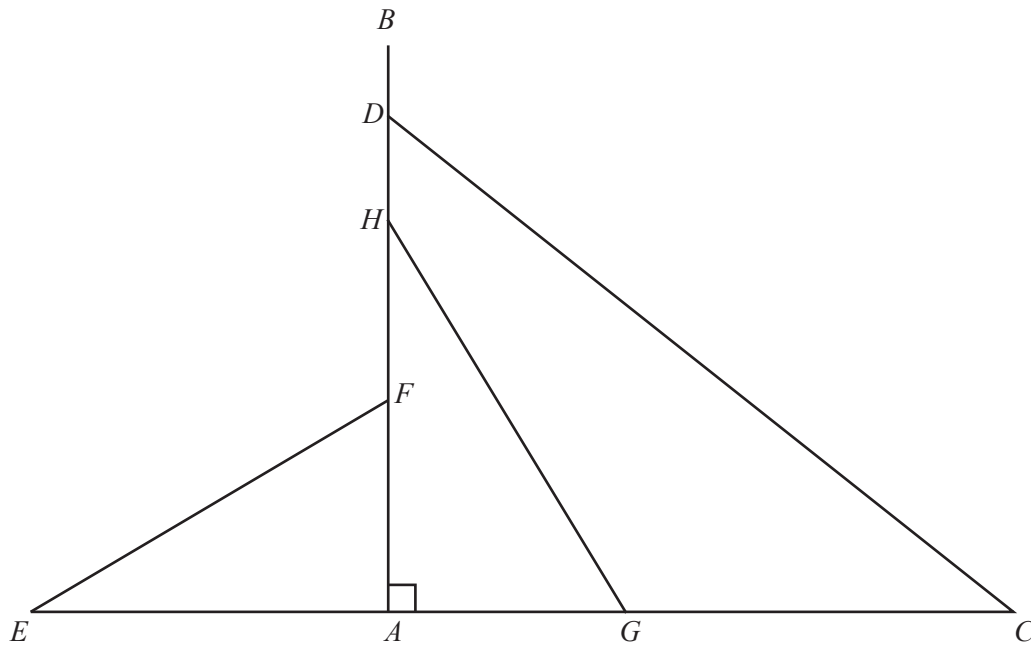
(c) Calculate AC .

Answer m [3]

(d) The perpendicular distance between the parallel lines CA and DF is 7 m.

Calculate the area of $ACDF$.

Answer m² [4]



The diagram shows a vertical radio mast, AB .

Three of the wires that hold the mast in place are attached to it at F , H and D .

The base A of the mast, and the ends E , G and C of the wires are in a straight line on horizontal ground.

- (a) The wire CD has length 65 m.
It is attached to the mast at D where $AD = 40$ m.

Calculate AC .

Answer m [2]

- (b) The wire EF makes an angle of 25° with the horizontal and is of length 30 m.

Calculate AF .

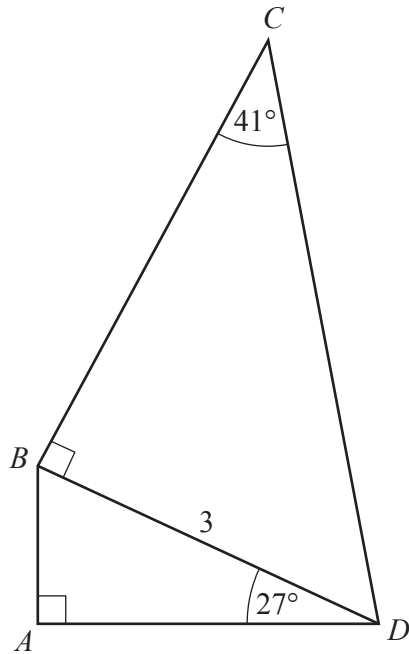
Answer m [2]

- (c) $AH = 35$ m.

The wire HG makes an angle of 30° with the mast AB .

Calculate HG .

Answer m [3]



In the framework $ABCD$, $BD = 3$ m.
 $\hat{BDA} = 27^\circ$, $\hat{BCD} = 41^\circ$. $\hat{DBC}$ and $\hat{DAB}$ are right angles.

(i) Find AD .

Answer m [2]

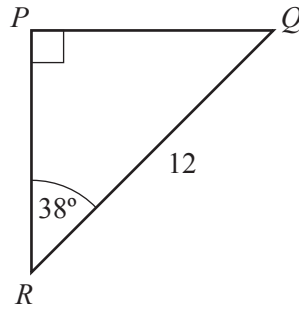
(ii) Find CD .

Answer m [3]

- (b) In triangle PQR , $PQ = 3$ m and $QR = 5$ m.
The area of triangle $PQR = 6 \text{ m}^2$.

Find the two possible values of $\hat{PQR}$.

Answer $\hat{PQR} = \dots\dots\dots$ or $\dots\dots\dots$ [3]



Triangle PQR has a right angle at P , angle $PRQ = 38^\circ$ and $RQ = 12$ cm.

(a) Calculate PQ .

Answer cm [2]

(b) S is a point such that angle PRS is a right angle and $QS = 10$ cm.

Calculate the two possible values of angle QSR .

Answer or [4]